Design and Fabrication Methodologies for Repurposing End of Life Metal via Robotic Incremental Sheet Metal Forming

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This paper investigates an integrative approach to robotic incremental sheet metal forming (RISF), which connects the registration of variable material properties and geometries to the re-forming of pre-made components beyond their initial formulations. Re-using rather than recycling metals can save the significant energy costs that come with having to melt, purify and re-manufacture products, as well as saving the costs of the new object it replaces. In this paper, we describe a workflow that connects 3d scanning, design automation and fabrication. The method goes beyond state of the art for RISF by challenging the assumption of starting from a flat unused sheet of metal, opening up the potential of RISF for material reuse. Our approach is demonstrated through the fabrication of a series of bench seating elements from oil drum geometries, however is generalisable to other input materials and output geometries. 3d scanning is used to register varying geometric features such as rolled beads, irregularities such as dents and holes, and material properties such as corrosion.

Keywords: *robotic fabrication, re-use, upcycling, incremental sheet metal forming*

INTRODUCTION

The growing impact of material consumption and waste on human health, economy and the natural ecosystem is motivating an increasingly urgent shift to resource efficient practices across architecture, construction and design. Central to this shift is the move from a linear to a closed-loop economy - a circular economy that reduces waste to a minimum via reusing, repairing and recycling.

Current research explores a variety of design and manufacturing-led approaches that aim to minimise waste, including design for disassembly, design for bio-materials, and design from reuse. In this paper we work with the concept of material reuse, which addresses existing products and places focus on extensions to their life cycle, and with a specific material focus on scrap steel. The production of steel is a highly energy-intensive process, and has maintained

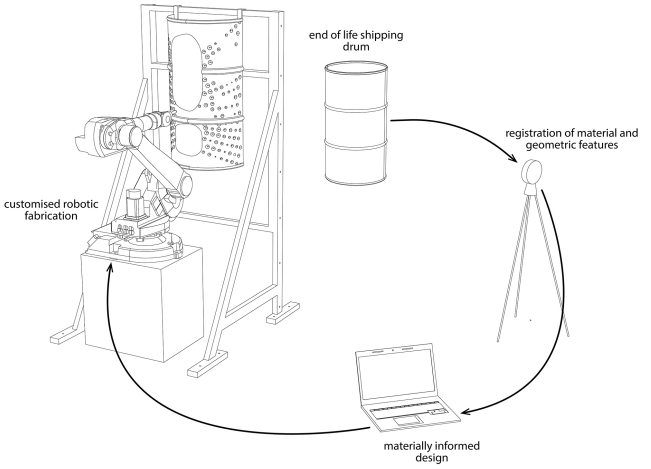


Figure 1
Registration, design and fabrication workflow

a relatively constant CO2 intensity over the past two decades [1]. Steel has the advantage of being 100% recyclable and scrap is used within both primary and secondary production processes, where it can result in up to 70 - 90% reduction in the energy cost for secondary production. However, secondary production is still highly energy intensive, much scrap steel not recycled, and large quantities of material are lost or deemed 'unrecoverable' with each use cycle and excluded in the calculation of recycling rates [2]. A lack of methods for processing reclaimed elements through automated fabrication is recogised as a barrier to reuse of this material [3].

Our paper introduces an innovative fabrication process - robotic incremental sheet metal forming (RISF) - and investigates how RISF can enable the direct reuse of scrap steel to make other products. Reusing rather than recycling metals can save the significant energy costs that come with having to melt, purify and re-manufacture products, as well as saving the costs of the new object it replaces. We demonstrate that as a tool for re-use, RISF opens up new possibilities for old scrap, and dramatically expands formal possibilities where no or only subtractive alterations are the norm. We firstly describe state of the art in architectural incremental sheet metal forming, the registration of relevant material properties, then detail our methodology for forming non-flat metal sheet into new products informed by local geometrical and material properties. We demonstrate this workflow (see figure 1) on the case of a bench seat, although the method is developed to be more widely applicable to other initial input objects and architectural cases. Lastly, our paper outlines the wider possibilities of this approach, which are found in re-use but also hybrid manufacturing scenarios, and the further research that is required.

FABRICATION PRACTICE AND PROPERTIES

Robotic Incremental Sheet Forming (RISF) is a variant of the flexible Incremental Sheet Forming (ISF) manufacturing process first developed in Japan in the 1980s and 1990s. A manipulative freeform process for imparting 3D form onto thin metal sheet, RISF moves a simple tool from either one or two sides causing a highly localized plastic deformation (Jeswiet et al 2005). A single sided approach typically utilizes one tool and a blank, while double sided forming provides further flexibility for forming out of plane in opposing directions. Where prior architectural research has investigated the structural and ex-

pressive potential of single sided (Bailly et al 2014, Nicholas et al 2016, Nicholas et al 2018) and double sided (Nicholas et al 2017) incremental forming, the starting geometries are always flat, new sheets of metal.

In this research, point cloud scanning is used to capture the geometry and surface properties of non-flat, non-new thin sheet metal objects. Within industry, structured light, time of flight and photogrammetric scanning methods are commonly used to measure the dimensional accuracy of parts and enable geometry and surface inspection, typically for the purpose of quality control (Nicholas et al 2020a) or for reverse engineering. In these applications, the captured 3d point cloud is used to create or update a 3d model from scan data of an unknown or uncertain physical part (Hale et al 2020, Brandt 2005). Recent research seeks to close the loop by connecting scanning to the re-manufacture or repair of damaged parts (Buican et al 2014), or the continuing robotic manufacture onto unknown geometries (Nicholas et al 2020b).

To form metal into a desired shape, the RISF process induces local strains, activating the material's capacity for plastic deformation under tensile stress. Corrosion, an environmentally driven process of chemical degradation, can negatively impact this capacity in two ways. Corrosion can be either local - 'pitting' - or uniform, which affects a larger area of the surface. Uniform corrosion can be simply calculated as a reduction of thickness, and has a minimal impact on the ultimate tensile strength. However corrosion also causes brittleness, with rusted areas of metal becoming less ductile and losing its ability to have its shape changed. This behaviour is more difficult to calculate but much more significant from the perspective of the fabrication process. Corrosion damage can be classified via visual examination. For example, in Sweden corrosion is classified from grades A (Steel surface covered completely with adherent mill scale and with little if any rust.) through grade D (Steel surface on which the mill scale has rusted away and on which considerable pitting is visible to the naked eye), while in Japan a 5 grade scale based on visual examination and measure of surface thickness is used. Research by Chun et al 2013 has aimed to standardise the Japanese scale, which they characterise as highly subjective and qualitative, through the application of image processing techniques and logistic regression. Similar research by Yan et al 2014 also uses machine vision and connects the same classification system to a machine learning-based approach. More recent research such as Petricca et al 2016 continues this approach, connecting image-based analysis to more powerful machine learning. This connection is a next step for our research, which in this paper is constrained to image-based analysis using pixel values. This approach is vulnerable to changes in lighting condition, where darker pixels can be mistakenly interpreted or compared as lighting conditions change.

METHODOLOGY

Our methodology connects the property registration of a found material with its further working. To summarise: 1. Shape and material properties are registered through 3d scanning, 2. Any zones of rust are classified into 3 material property parameters, 3. Additional geometry to be formed into the base geometry is introduced in a CAD environment, 4. A structural analysis is performed to generate a rigdisation pattern for areas with high von mises stress, 5. Robot toolpaths are extracted from the model, and 6. New geometries are formed into the found geometry. These steps are expanded upon in the sections below.

Registration

3D scanning is performed with Intel's RealSense LiDAR scanner. The found metal object is scanned in the forming frame. An anti reflective spray is applied to the metal to improve the quality of the scan and reduce noise caused by reflectivity. A toolkit was developed to record and read the pointclouds (see figure 2) in Rhino3D and Grasshopper, software used further for the digital workflow. The scanner can also

be used to register the fabrication process itself. By recording frames at a certain time interval, the fabrication process can be optimized by analyzing the recorded mesh against the model in the CAD environment to measure real time forming tolerances.

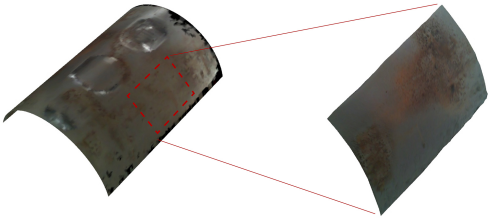


Figure 2
Registration of material properties

Processing of material properties

The pointcloud provides geometric information for the found metal object including any dents, beads, holes etc. Material properties impacting the surface quality and structural properties are extracted via analysis of the colour of each point within the 3d scan. Working within an HSL colour space, colours associated with rust are identified through their luminance values . As these are variable according to light levels, these values are determined on a case by case basis.

All points within the pointcloud are then categorised into one of three different property levels. Level one means that it is almost rust-free and the forming process can be fully executed. Level two evidences a corrosion process from a low to medium extent and forming is only possible to a limited wall angle and depth. Level 3 is classified as advanced corrosion in which the material has no stability and is of a brittleness that makes it unsuitable for forming.

Placement and sizing of new geometries

The design workflow processes two different models in parallel - nurbs and mesh based. The entire 3d model is built as a Nurbs model and all geometric manipulation takes place in this model. However, colour information is obtained from the mesh vertices, and a structural analysis is performed on the mesh.

The geometry to be formed is manually positioned on the Nurbs model of the found object. First, an over-sized region of interest is selected in which the geometry should be placed, for example only in the upper area of the original object. The selection of this area is however only approximate, so that the position of the geometry can be adjusted to account for local material properties. In order to adjust this positioning, all vertices of the mesh that lie within the region of interest are interrogated for their level of corrosion, with visual feedback given to the designer. The most important criterion is level 3 corrosion, which is not formable. Once positioned, the Nurbs model is converted into a mesh to perform a structural analysis. The grasshopper plugin Karamba3D is used for structural analysis. Depending on the case, different structural information can be output, including the magnitudes and directions of principal forces. This output introduces the possibility to generate new geometries informed by structural feedback. After the final geometry has been defined, the print path is generated.

Robotic fabrication setup

The method has been developed and tested on two robotic setups - a UR5e and an ABB IRB 1600. Initial testing of the fabrication process explored RISF on non-flat metal sheets. The setup for this experiment uses a UR5e robotic arm with an 18 mm diameter forming tool operating with a speed of 45 mm/s from the convex side of the curved sheet. The tool forms a 0.3 mm aluminum sheet fixed in a jig that clamps it to the shape of a semi-cylinder (see figure 3). The target geometry is half of an ellipsoid. The second stage of the investigation used a ABB IR160 robotic arm with an 18 mm diameter forming tool operating with a speed of 32 mm/s. The holding jig includes three rigid belts which clamp a 0.5 mm mild steel sheet in the shape of a semi-cylinder (see figure 4). These initial tests showed that non flat forming was possible, however that additional strategies were required to prevent buckling.

Figure 3
UR5e fabrication setup

Figure 4
ABB fabrication setup

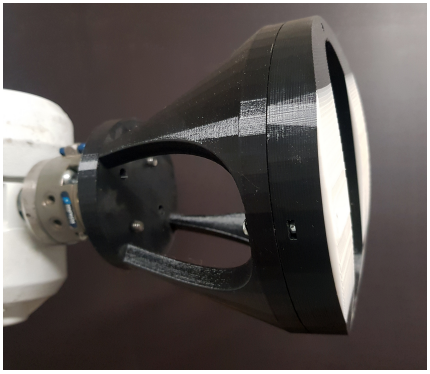
Figure 5
First design of holder tool with interchangeable head

ROBOTIC STRATEGY TO COUNTERACT BUCKLING

In order to counteract buckling, our initial strategy is to use a 3D printed holder tool attached to a second robot, which would locally support the sheet in the forming area from the concave side (see figures 5-7). The holder tool has two components, one which is fixed to the robot arm and an interchangeable head which varies with the size and shape of the formed geometry (see figure 5). The curvature of the edge touching the metal sheet is the same as the curvature given by the rig. This stops the sheet from buckling, however it also limits the freedom of design of the geometry as a new head would need to be printed for each different geometry. A second design of this tool consists of a single component tool with four supporting points (see figure 6). This upgrade allows for more freedom of shape, but it is limited in its ability to form larger areas.

Rig-based strategy

Our second strategy is the redesign of the rig further used in the second setup (see figure 8). The curved sheet is supported at top and bottom by metal rigid belts. Two intermediate belts are integrated with flexible placement according to the forming location. This introduces intermediate clamping that reduces the distance spanned by the metal sheet reducing the possibility of macro scale buckling of the barrel.



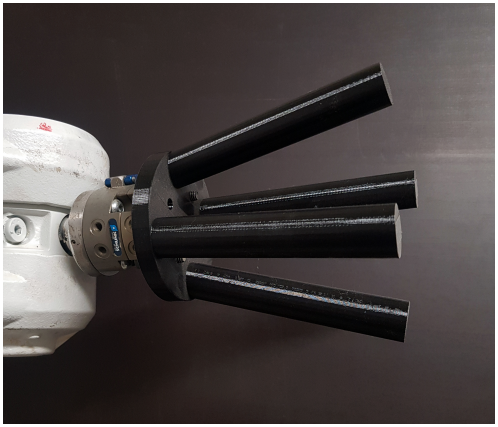


Figure 6
Second design of holder tool with four supporting points

Toolpath-based strategy

The fabrication toolpath is created by contouring the geometry in the CAD model and spiralizing the curves for continuous movement. In the case of larger irregular geometries, it was found that the metal sheet tends to pillow where the distance from center to edge is highest. Our strategy to reduce this tendency is to introduce a cone perimeter encasing the initial geometry. This perimeter is formed first and the toolpath progressively transitions to the target geometry. In addition, the metal is pressed into the geometry every 15 layers (see figure 9). The metal is thus better controlled and takes on the desired shape.

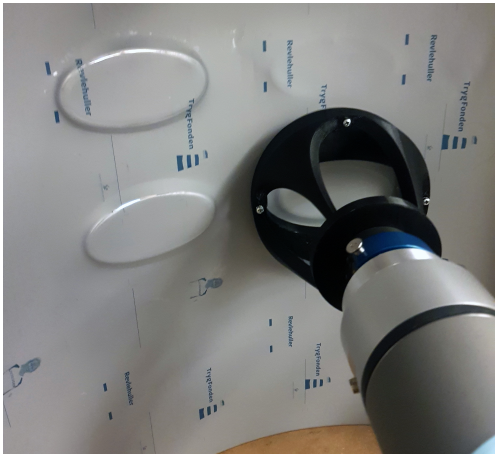


Figure 7
Forming using local support tool (back view)

INCORPORATING MICRO BUCKLING

The largest geometry is formed first. In our testing, it is typically positioned near the center of the metal sheet. The forming of this feature increases the stiffness of the entire sheet. Secondary geometries are then formed: rigidisation pattern consisting of cone-shaped geometries informed by the structural analysis from the digital workflow. Micro buckling appears within the rigidsation pattern when the forming area is already tensed. The design language of the micro-buckling can be changed by following certain rules such as the forming order of the cones, the positioning of the cones on the metal sheet, the proximity between them and their geometrical aspect such as base diameter and depth. For instance, when the cones are positioned on a linear or concentric grid the buckling seam tends to follow a line-shape in between the cones. Whereas when the cones are positioned in a diagrid, the buckling seam follows the grid pattern. By controlling the proximity between the cones and their size we can change the depth of the buckling seam. Therefore, when cones are closer and larger, the seam is more prominent, whereas when the geometries are further from each other and smaller there it fades away.

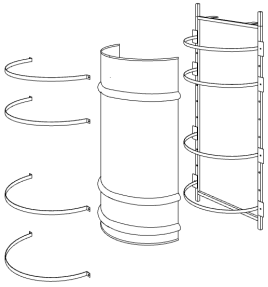


Figure 8
Metal rig with intermediate rigid belts

Figure 9
Fabrication toolpath strategy

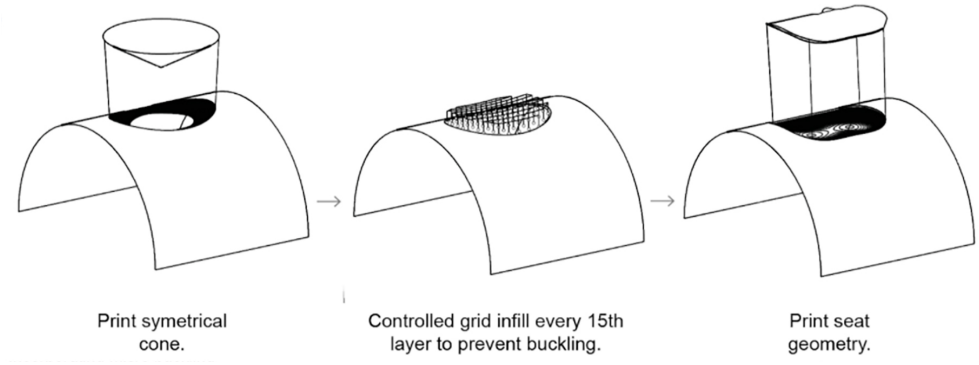


Figure 10
Deliberately induced micro-buckling (back view)

CASE STUDY - DESIGN FOR A BENCH

This methodology is tested through a case study, to design and fabricate a bench seat from a barrel which reached its end of life. In this paper, the digital development and initial prototyping of this case study is emphasised. Our design speculation consists of two scenarios, using a 55-gallon and a 100-gallon standard barrels. In our digital model, the barrels are created using standard dimensions and bead positioning, whereas the rust pattern has been generated using noise patterns in order to imitate randomness of weathering. The classification of the material properties are divided in 3 parts according to the method described above: no rust, light rust, which allows for limited forming and heavy rust where no alterations can be made. We assume that the bead rolls on the 55-gallon barrel are formable, since they are less than 10 mm in depth, whereas the ones on the 100-gallon barrel are more prominent, hence not formable. Two different seat geometries are used, adult size and children size. These geometries design are fixed in order to follow the ergonomic aspect of a chair. Tubular legs are attached to the reinforcing beads of the barrel.

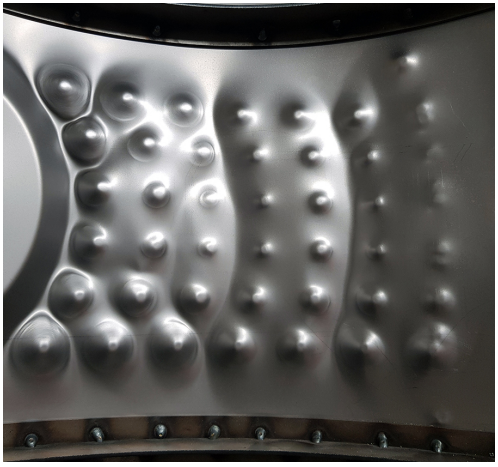


Figure 11
Deliberately induced micro-buckling (front view)

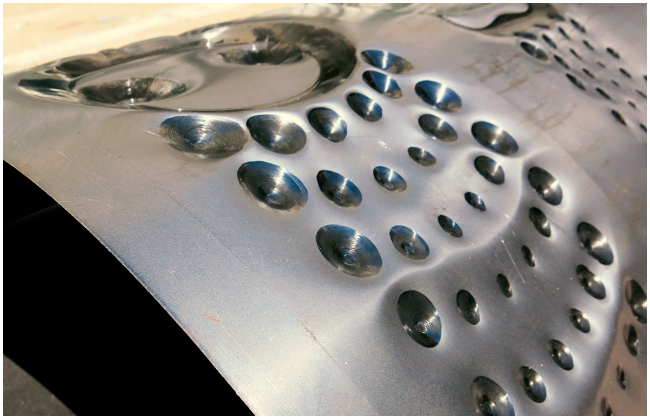


Figure 12
Formed rigidisation geometries

The methodology described above is applied to this setup to generate a design geometry and robotic toolpaths including the initial properties, the seat geometry and a rigidisation pattern (see figures 10 - 12). In the case of the 55-gallon barrel (see figure 14), two children-sized seats are formed according to an analysis of the existing material and geometric aspects. In the case of the 100-gallons barrel (see figure 15), one adult-size bench is formed, however a smaller size could be a second iteration for the same rust pattern.

Inputs to the structural analysis include the load of an adult (80kg) or child (30kg) distributed equally over the seat geometry, restraints at the leg positions, and material specifications defining the thickness of the material. L1 = 1mm, L2 = 0.5mm, and L3 = 0.01mm. The determination of the seat position and the feet allows an exact structural analysis of the force transmission to the geometry.

In the next step, the pattern is generated which is divided into several operations. First, an unequal grid is generated in the Nurbs model between the seat and the edge of the barrel. Cones positioned on this grid are sized according to the structural model, where the force in each nearest vertice is calculated. This model generates a specific force transfer on each position which we use to determine the radius and depth of a cone of the pattern. The cones act to stiffen and rigidify the thin metal skin under loading - a baseline comparison shows that a half-barrel geometry without cones has 43% performance under a sitting load compared to the same geometry

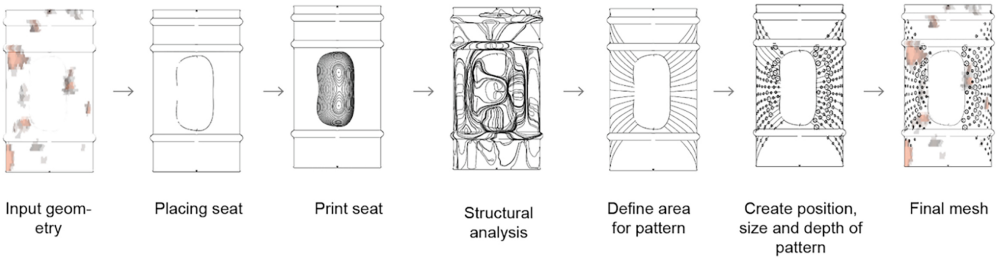


Figure 13
Workflow for design for a bench

with the determined distribution of cones. Next, the position of the cones is reconfigured according to the rust classification. If a cone is located in an unformable area (L3), it is eliminated, whereas if a cone is on a constrained deformable area (L2), its size is reduced. Their position further depends on the reach of the robot arm, since in our case, the cones cannot be formed within 60 mm from the edge of the half-cylinder. The size and depth of each geometry is thus determined before the final nurbs model is generated, from where the toolpath is extracted. It is then converted back into a mesh to be colourised for visual purposes and, if necessary, a second structural analysis is performed. Toolpaths are extracted and fabrication achieved using the toolpath-based strategy described above.

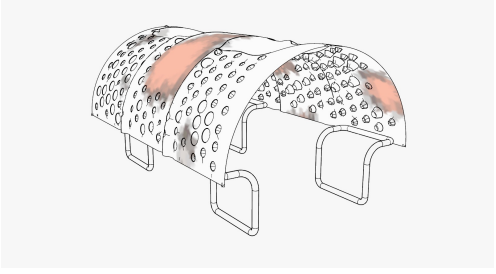


Figure 14
Case study examples - 55 gallon barrel

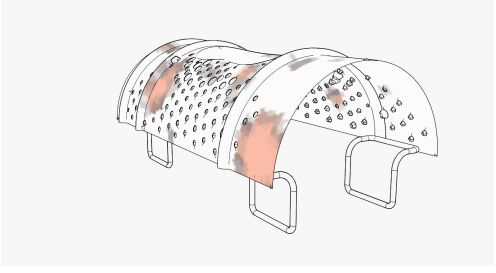


Figure 15
Case study examples - 100 gallon barrel

DISCUSSION AND CONCLUSION

This paper has introduced a workflow that connects 3d scanning, design automation and robotic fabrication. The workflow aims to support the direct reuse of scrap metal through incremental sheet metal forming, avoiding the energy cost of recycling. This approach is demonstrated through the design and prototyping of bench seating elements from oil drum geometries. 3d scanning is used to enable incremental forming with non flat geometries and partially corroded materials. The forming geometries aim to increase the structural performance of the oil drum, while moving away from a symmetrical geometry in order to integrate the ergonomic functionality of a seat. Here, our vision-based approach to incorporating material properties has established a basic pipeline for information from the found object into the digital representation, and further work is planned to incorporate machine learning based models to categorise corrosion. The localised support tools and toolpath based methods developed to avoid buckling are still in development, but our use of them in these experiments has shown that both approaches could be needed to form with non flat, non-tensioned sheet metal. Further work will focus on testing more varied metal objects through the workflow, on developing further iterations of the support tool that enable even greater forming flexibility, and altering the toolpath approach to better avoid pillowing during the forming of deep geometries by moving iteratively outwards and deeper through multiple passes until the target depth is reached. Finally, in the context of material reuse, our workflow could support highly localised reuse and remanufacturing cycles. These methods could be expanded into a larger scale assembly of re-used metal sheets across a variety of architectural and construction applications, drawing on different material streams. Many potential input sources come in standardized sizes such as household appliances (fridge, dishwashers, washing machines, etc.), spiral or square duct vents, corrugated sheets, or car/airplane skin. Applications could include façade cladding, screens or canopies. Where we have worked here with single objects, in next steps formed panels will be assembled into free-standing spatial structures with the forming process used to embed additional functionalities related to water flow, reflection or shading. In more conventional applications, the developing methods could

also enable hybrid manufacturing processes in which initial forming was achieved via traditional metal forming techniques such as stamping or deep drawing, to which subsequent detailed forming is then applied. This approach would connect the speed of instantaneous forming processes to the customisation achievable via incremental sheet metal forming.

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